1 Bilinear Forms

Definition 1.1 (Bilinear form)**.** Let V be a vector space over F. A *bilinear form* on V is a function $B : V \times V \to F$ which is linear in each variable separately when the other variable is held constant. That is, for all $v_1, v_2, v \in V$ and $\alpha \in F$, we have

$$B(\alpha v_1 + v_2, v) = \alpha B(v_1, v) + B(v_2, v)$$

and

$$B(v, \alpha v_1 + v_2) = \alpha B(v, v_1) + B(v, v_2).$$

Exercise 1.2. Show that $B : F^2 \times F^2 \to F$ defined by

$$B((u,v),(s,t)) = ut - vs$$

is a bilinear form on F^2.

Exercise 1.3. Let $V = M^{n\times 1}(F)$. Show that if A is an $n \times n$ matrix, then

$$B_A(x,y) = x^T Ay$$

is a bilinear form. Note that the bilinear form of the previous exercise is of this form with $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$.

Exercise 1.4. Show that $B : \mathscr{P}_n(\mathbb{R}) \times \mathscr{P}_n(\mathbb{R}) \to \mathbb{R}$ defined by $B(p,q) = p(2)q'(3)$ is a bilinear form on $\mathscr{P}_n(\mathbb{R})$.

Exercise 1.5. Let V be any vector space over F and fix $f, g \in V^*$. Show that $B : V \times V \to F$ defined by $B(u,v) = f(u)g(v)$ is a bilinear form on V. More generally, if $f_1, \ldots, f_n, g_1, \ldots, g_n \in V^*$, then $B(u,v) = \sum_{i=1}^n f_i(u)g_i(v)$ is a bilinear form on V.

Remark 1.6. Note that $V \times V$ is a vector space with respect to the operations

$$(u_1, v_1) + (u_2, v_2) := (u_1 + u_2, v_1 + v_2)$$
$$\alpha(u,v) := (\alpha u, \alpha v).$$

If $B : V \times V \to F$ is a bilinear map on V, we can then ask under what condition is B also a linear functional on $V \times V$? For such a B, we would have $B(\alpha u, \alpha v) = \alpha B(u,v)$ since B is linear as well as $B(\alpha u, \alpha v) = \alpha^2 B(u,v)$ since B is bilinear, which implies $(\alpha^2 - \alpha)B(u,v) = 0$ for all $u, v \in V$ and all $\alpha \in F$. For this to hold for $\alpha \neq 0, 1$, we must therefore have $B(u,v) = 0$ for all $u, v \in V$, hence B a bilinear map on V is also a linear functional on $V \times V$ if and only if B is the zero map (which is easily verified to be a bilinear form on V).

Exercise 1.7. Let $V^{(2)}$ denote the set of all bilinear forms on V. Show that this is a subspace of $F^{V\times V}$, the vector space of all maps $V \times V \to F$.

Suppose now that V is a finite-dimensional vector space with $\dim V = n$. We will now prove that $V^{(2)}$ is also finite-dimensional with $\dim V^{(2)} = n^2$.

Definition 1.8 (Matrix of a bilinear form)**.** Let B be a bilinear form on V and let $(e_1, \ldots, e_n)$ be an ordered basis of V. Define the *matrix of* B with respect to this ordered basis to be the matrix $[B]$ whose ij-entry is given by

$$[B]_{ij} = B(v_i, v_j).$$

Exercise 1.9. Find the matrix of the bilinear form $B : \mathscr{P}_2(\mathbb{R}) \times \mathscr{P}_2(\mathbb{R}) \to \mathbb{R}$ of Exercise 1.4, defined by $B(p,q) = p(2)q'(3)$ with respect to the standard ordered basis of $\mathscr{P}_2(\mathbb{R})$.

Exercise 1.10. Prove that the map $T : V^{(2)} \to M^{n\times n}(F)$ defined by $T(B) = [B]$ is an isomorphism. This proves that $V^{(2)}$ is finite-dimensional with $\dim V^{(2)} = n^2 = (\dim V)^2$.